DSML GROUP PROJECT REPORT · MBA ANALYTICS 2025-27

Predicting Regions With High Electric Vehicle Adoption in India

Predictive Analytics for Charging-Infrastructure Planning

Indian Institute of Management Kashipur, Uttarakhand

GROUP 11

- Abhijay Kumar Jayant · MBAA25004
- Bhavika Garua · MBAA25017
- Shruthi G Bhosle · MBAA25160
- Vaidehee Ghorpade · MBAA25073
- Anjali · MBAA25008
- Nandini Talwar · MBAA25035
- Jeptha H Lalawmpuia · MBAA25024

EXECUTIVE SUMMARY

Registration Data Signals Where EV Adoption Is Heading

418,372	34	6.85%	0.92	8
Registration records analysed (2019 – May 2024)	States & union territories scored	National EV adoption rate, Jan–May 2024	Correlation: 2023 vs 2024 adoption rate	Regions on the final consensus shortlist

India's EV registrations are growing far faster than its charging infrastructure and adoption is spreading unevenly across states. This project asks a narrow, useful question: **using only public vehicle-registration data, can we identify in advance which states are heading toward high EV adoption?**

- All four classifiers (Logistic Regression, Decision Tree, Random Forest, Gradient Boosting) reached 1.00 accuracy / ROC-AUC because 2023 adoption share alone separates the classes almost perfectly. We treat this as a warning about sample size (n=34), not an achievement.
- Regression estimates of the exact 2024 adoption rate are far more modest (best $R^2 = 0.67$), showing the models can place a region in the right group but not pinpoint its exact share.
- Adoption rate and adoption volume are nearly independent Goa leads on % (16.44%) with under 6,000 EVs, while Uttar Pradesh's 8.26% represents over 130,000 EVs. Planning must use both, never rate alone.
- A data-quality problem was uncovered: 342 of 418,372 rows (0.08%) account for 69% of all registrations, materially distorting Rajasthan's ranking.

IIM KASHIPUR

Why Reactive Charging Infrastructure Fails Fast-Growing Markets

THE PROBLEM

Charging infrastructure in India is largely deployed **reactively** - investment follows visible EV concentration, so the regions that most need early support are often the last to receive it.

Question 1 - Classification: Can a region's registration history predict whether it lands in the top quartile of adoption next year?

Question 2 - Regression: Can the same history estimate the exact adoption rate as a continuous number?

Diagnostic question: Which historical signals actually carry the information level, growth, or volume?

KEY OBJECTIVES

1. Convert office-level monthly records into a state-level EV adoption history
2. Engineer explanatory features using only pre-2024 information (no leakage)
3. Define a defensible high-adoption target and build classification models
4. Estimate the continuous 2024 adoption rate via regression models
5. Identify the historical indicators the models actually rely on
6. Segment regions into adoption profiles using K-Means + PCA
7. Produce a scored, planning-ready shortlist of priority regions
8. Honestly assess data limitations before any operational use

DATASET DESCRIPTION

From 418K Office-Level Rows to a 34-Region Analytical Table

Property	Value
Rows (observations)	418,372
Date coverage	Jan 2019 – May 2024
States / union territories	34
Registration offices	1,414 (names)
Distinct fuel types	22
Missing values / duplicates	None
Total registrations recorded	771,822,033

EV Definition: fuel_type contains "Electric"

PREPROCESSING PIPELINE

1. Office-month-fuel rows (418,372)
2. Flag EV rows & sum by state-year
3. Compute adoption rate = EV ÷ Total regs
4. 1 row per region, years as columns
5. Engineer 11 features from ≤2023 data only
6. **34-region modelling table + 2024 target**

DATA QUALITY CHECK

A Small Number of Rows Distort the National Totals

342 / 418,372

Rows above 500,000 registrations (0.08%)

69%

Share of total registration volume in those rows

~90%

Of Rajasthan's 2024 EV total from one office

6.80% → ~20%

Rajasthan's adoption rate before / after capping

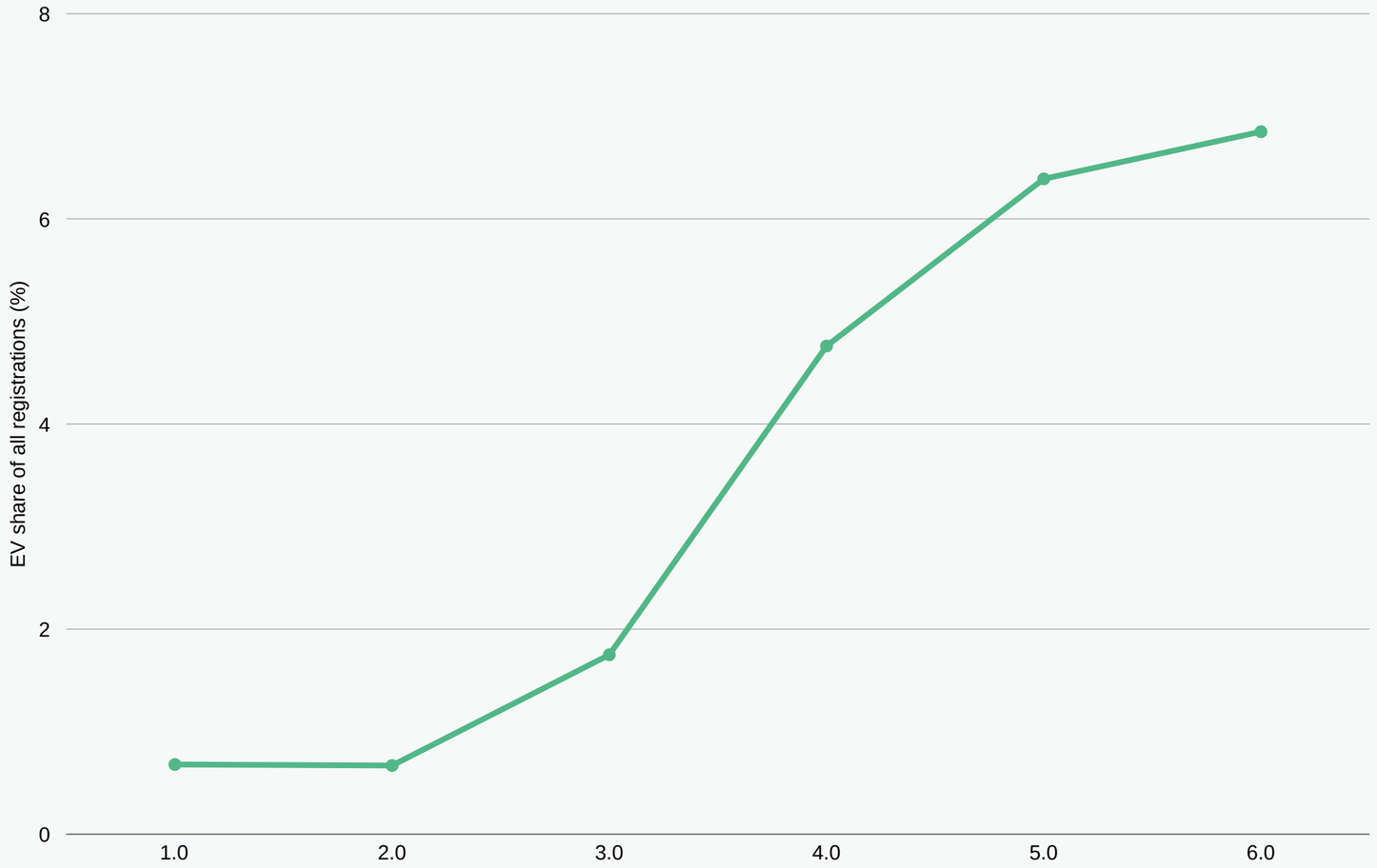
OFFICES CONTRIBUTING THE LARGEST SHARE OF EXTREME ROWS

Office	State	Rows > 500K	Registrations in those rows
Lower Siang	Arunachal Pradesh	46	66,569,959
Narasaraopet RTA	Andhra Pradesh	35	64,906,570
Hindupur RTA	Andhra Pradesh	35	64,102,492
Sikar Vehicle Fitness Center	Rajasthan	41	59,109,105
Nilambur SRTO	Kerala	34	48,428,083

Why it matters: Fitness-testing centres (re-certifying vehicles already on the road) are being counted as new registrations. National 2020 totals of 388M registrations vs. 24.7M in 2019 show the same fingerprint. Rajasthan and Arunachal Pradesh results should be read with this caveat; results in this deck are a directional shortlist, not a settled ranking.

IIM KASHIPUR

National EV Adoption Rate Has Grown 10x Since 2019



WHAT IT SHOWS

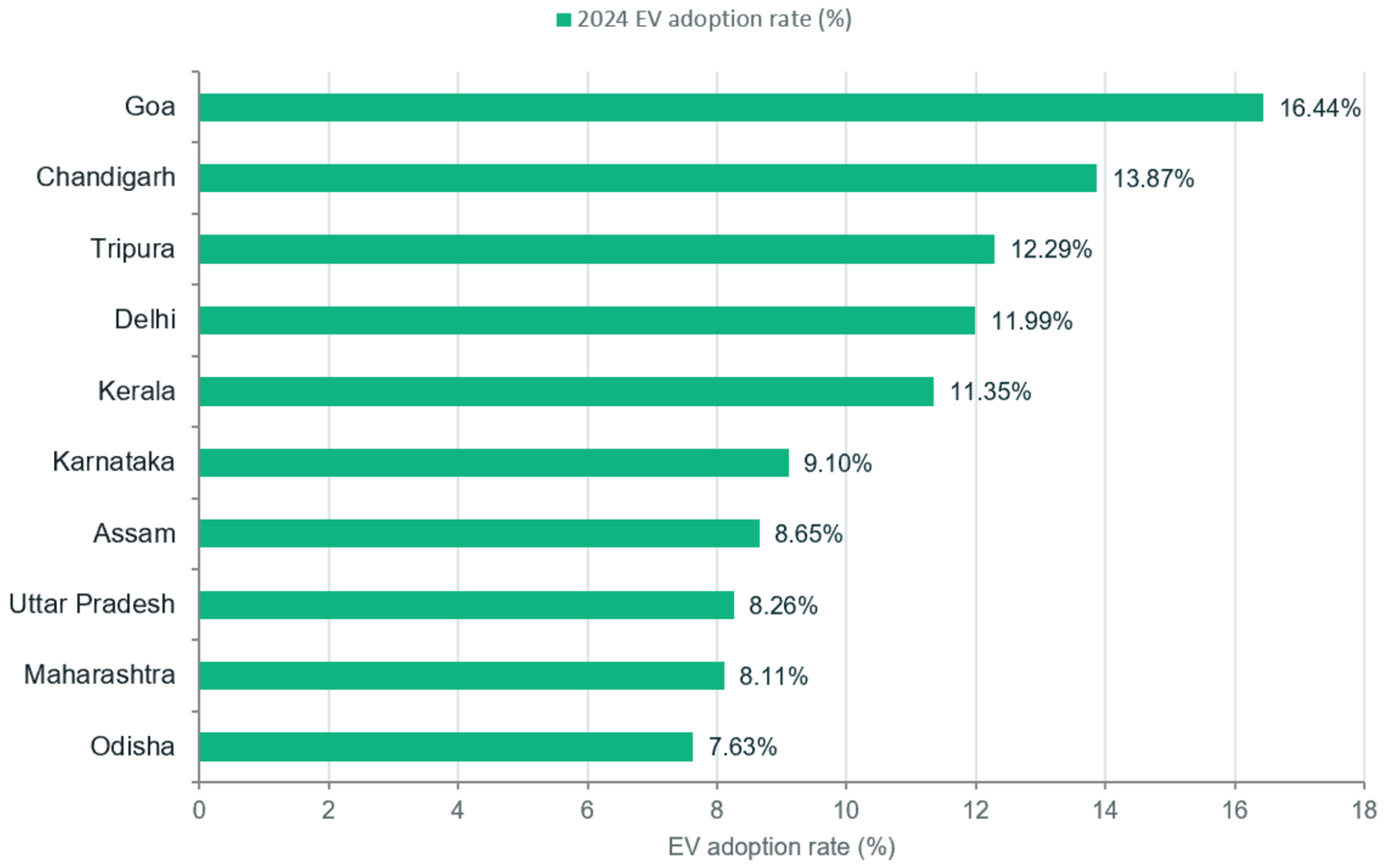
Flat at ~0.68% through 2020, then a sharp rise from 2021 the rate roughly triples between 2021 and 2022.

2024 covers only Jan–May, so the 6.39% → 6.85% step is not a genuine slowdown it is not a like-for-like annual comparison.

A rising national trend is a precondition for the whole project: if EV share were flat, next year's ranking would just be this year's ranking.

EXPLORATORY ANALYSIS · REGIONAL LEADERS

Goa leads at 16.4%, but the top 15 decline gradually, not sharply



75th-percentile threshold = 7.99%

9 regions qualify as "high adoption", 25 do not. Maharashtra (8.11%) and Odisha (7.63%) sit less than half a point apart, yet fall on opposite sides of the label.

Three region profiles emerge

Compact & urban: Goa, Chandigarh, Delhi — short trips, dense form

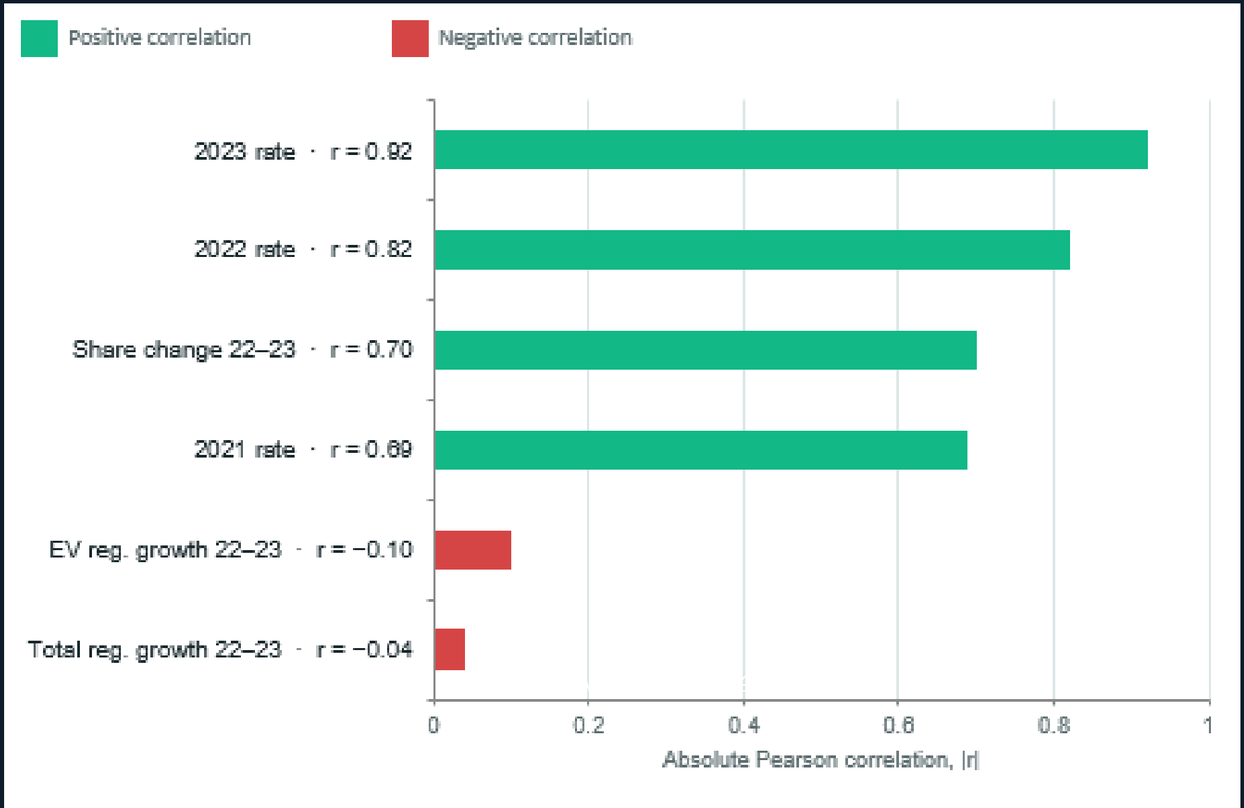
Policy-driven: Kerala, Karnataka — EV incentives, transit electrification

Harder to explain: Tripura, Assam — likely two/three-wheeler led

A gradual decline (not a sharp break) means the top-quartile cut is a reasonable but somewhat arbitrary line — not a natural break in the data.

7

Recent Adoption Level Predicts Next Year Growth Rates Don't



MULTICOLLINEARITY WARNING

Adoption is highly persistent: r = 0.69, 0.82 and 0.92 for 2021, 2022 and 2023 respectively recency matters.

The 2022–23 change in share (r = 0.70) also carries real signal but percentage growth in raw EV counts (r = –0.10) does not. Growth off a tiny base (40 → 160 EVs = 300% "growth") rewards regions that add nothing to national demand.

The three level variables also correlate with *each other* at 0.81–0.93, meaning they carry heavily overlapping information not three independent signals. This is why Logistic Regression coefficients must be read as a group, not individually.

IIM KASHIPUR

Eleven Features Built Strictly From Pre-2024 Information

Design rule: nothing from the 2024 outcome period enters the feature set every explanatory variable is dated 2023 or earlier, so the test is a genuine forward prediction.

Feature	What it captures
ev_adoption_rate_2021 / 2022 / 2023	Level of adoption in each individual year, so recency can be weighted
ev_share_avg_2021_23	3-year average, smooths one-off swings in small regions
ev_share_change_22_23	Momentum measured as a percentage-point difference (comparable across regions)
ev_reg_growth_22_23 / ev_cagr_21_23	Growth in absolute EV counts intended to capture acceleration
ev_registrations_2023 / total_registrations_2023	Absolute market size, since a share alone ignores volume
office_count_2023	Proxy for administrative reach / reporting completeness

DELIBERATELY EXCLUDED

id, category
— no variance / no information

state_code, office_code
— arbitrary identifiers

2019 / 2020 adoption rates
— near-zero for most regions, adds correlated noise

No formal feature selection was run up front with 34 rows and 11 features the ratio is tight. Feature importance (Slide 12) later shows 3 of 11 features carry ~95% of the signal, effectively performing the selection after the fact.

ML METHODOLOGY

Two Supervised Tracks, One Feature Matrix, Careful Validation

Classification

Binary: high_adoption (0/1)

Answers the planning question directly does this region belong on the shortlist?

Regression

Continuous: 2024 adoption rate

Gives a numeric estimate so regions can be ranked, not just grouped

Clustering

Unsupervised : K-Means + PCA

Groups regions into adoption profiles for portfolio-level planning

TRAIN / TEST DESIGN

75 / 25 split, fixed random seed → 25 training regions, 9 test regions

Classification is stratified: training = 18 normal / 7 high; test = 7 normal / 2 high.

5-fold stratified cross-validation (ROC-AUC) re-checks results across all 34 regions, since a 9-region test set alone is far too small to trust on its own.

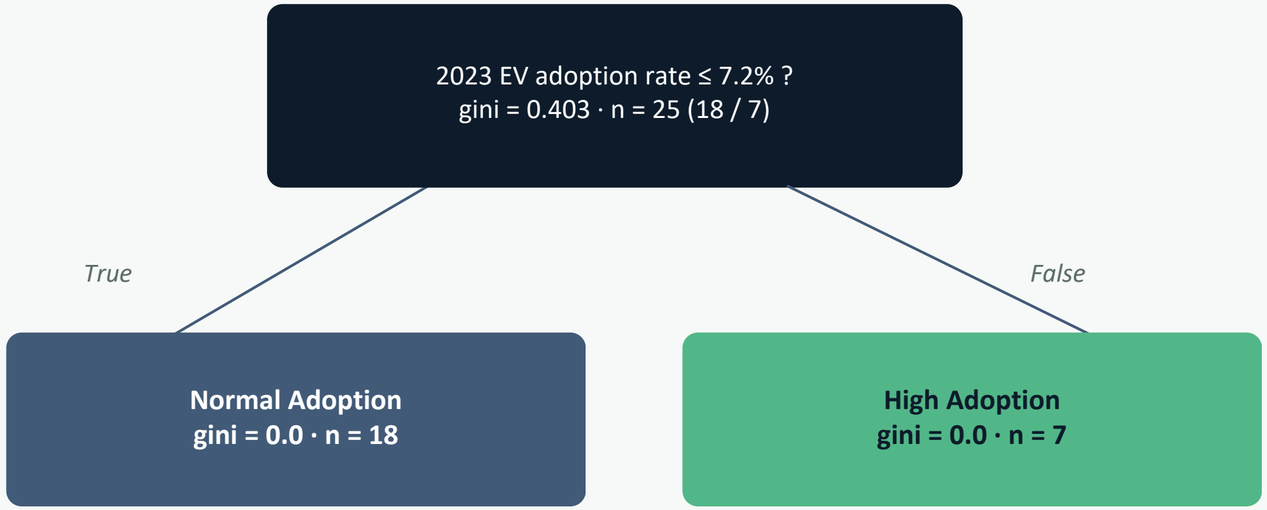
MODEL CONFIGURATIONS

Model	Key configuration
Logistic Regression	Standardised, L2 penalty, max 1,000 iterations
Decision Tree	Max depth 3, min 2 samples/leaf
Random Forest	300 trees, 70% features/split, min 2 samples/leaf
Gradient Boosting	150 stages, depth 2, learning rate 0.05
K-Means	k = 2–6 tested, 20 inits, chosen by silhouette

IIM KASHIPUR

One Threshold Separates Every Region Perfectly

THE FITTED DECISION TREE



The tree was allowed depth 3 and chose depth 1. In plain language: if a region's EV share exceeded ~7% in 2023, it was in the top quartile in 2024, every single time in this sample. The two test-set positives (Kerala 9.99%, Karnataka 8.74%) both exceed the highest test-set negative (Rajasthan, 6.36%) so any threshold in between classifies all 9 test regions correctly.

CLASSIFIER COMPARISON

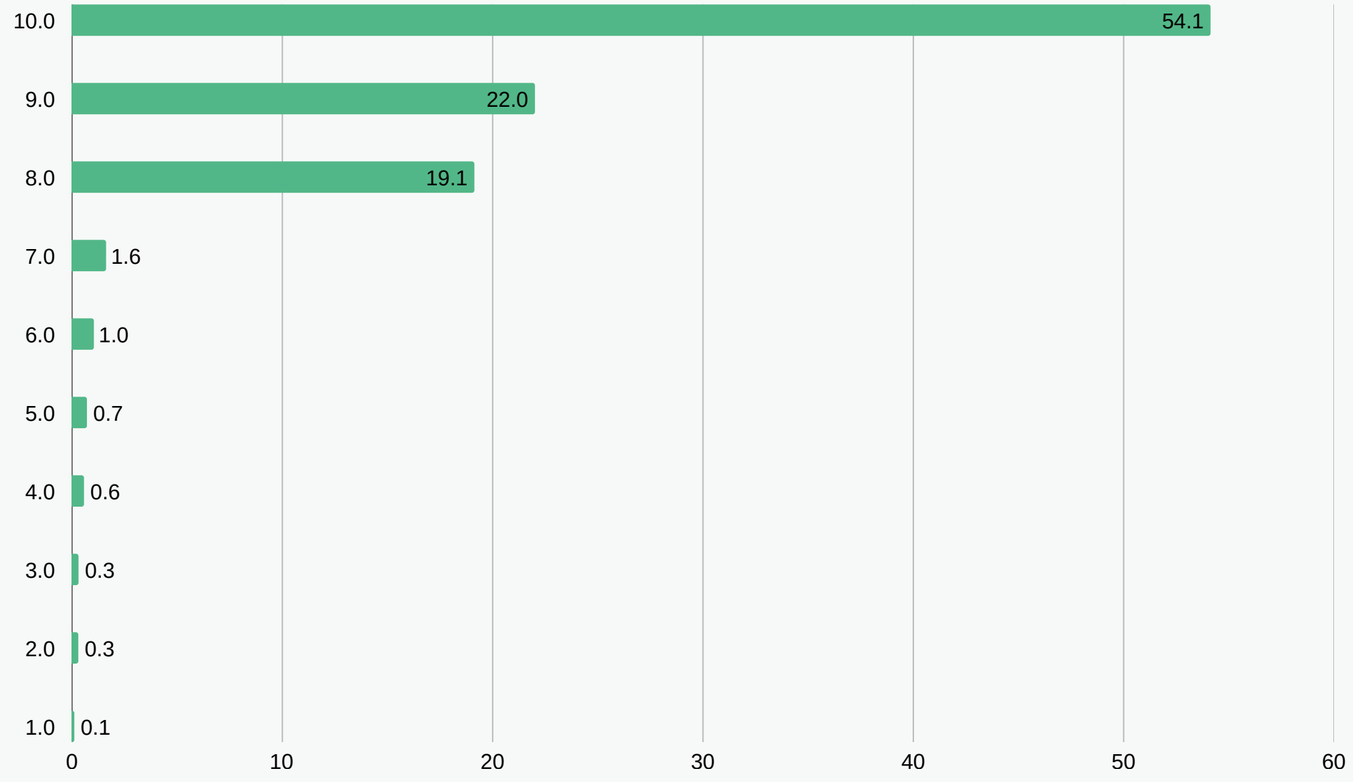
Model	Test Acc.	Test AUC	CV Mean AUC	CV Std
Logistic Regression	1.00	1.00	1.00	0.00
Decision Tree	1.00	1.00	0.98	0.04
Random Forest	1.00	1.00	1.00	0.00
Gradient Boosting	1.00	1.00	1.00	0.00

READING PERFECT SCORES HONESTLY

Four different algorithms reaching an identical ceiling is the signature of an easy, separable problem **not evidence of four excellent models.** With only 9 test regions (2 positive), a single misclassification would swing accuracy by 11 points.

The Decision Tree's 0.04 CV standard deviation (dropping to 0.90 in one fold) is the tell a hard single threshold is sensitive to exactly which regions land in each fold.

Three Features Carry 95% of the Random Forest's Decisions



Random Forest feature importance for high-adoption classification (share of total, %). The top three adoption-level features account for 95.3% of all importance.

REGRESSION: ESTIMATING THE EXACT RATE

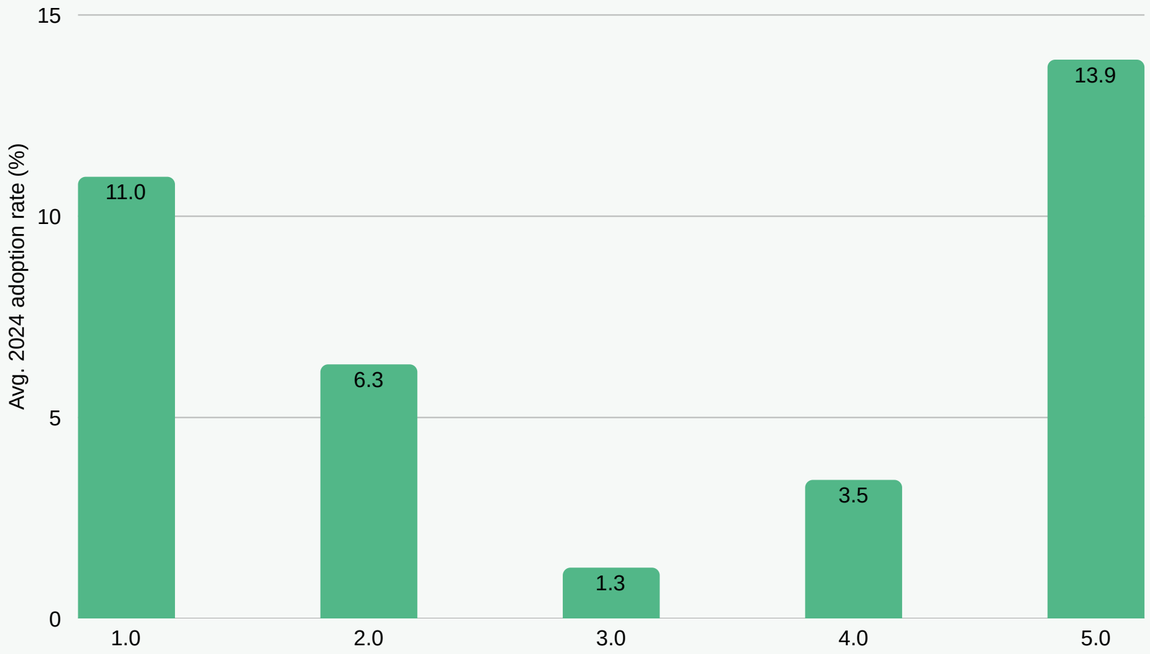
Model	R^2	RMSE	MAE
Random Forest	0.666	0.0283	0.0222
Gradient Boosting	0.607	0.0307	0.0245
Linear Regression	0.599	0.0310	0.0211

Best R^2 = 0.666 with a typical error (RMSE) of 2.8 percentage points, against a target with a 4.2-point standard deviation. Models can place a region in the right neighbourhood, *not on the right street.*

Linear Regression predicted impossible negative rates for Rajasthan and Sikkim (clipped to zero) a sign it fits the extremes poorly.

Regression is weaker than classification not by contradiction: sorting into two groups either side of one threshold is a far easier task than pinpointing a continuous value.

K-Means Reveals Five Distinct Adoption Profiles



Average 2024 adoption rate by cluster (n shown on next panel). Best k = 5 by silhouette score (0.478) — though scores for k = 3, 5, 6 are within 0.007 of each other, so the choice is only weakly supported statistically.

CLUSTER PROFILES

Cluster	n	Character	Example regions
0	3	Established high adoption	Tripura, Delhi, Assam
1	16	Broad middle market	Karnataka, UP, Maharashtra, Bihar…
2	10	Low adoption	West Bengal, Mizoram, Sikkim…
3	2	**Volume outliers (data artefact)**	Rajasthan, Arunachal Pradesh
4	3	Fast-accelerating adoption	Goa, Chandigarh, Kerala

Cluster 3 is an artefact, not a behaviour segment: Rajasthan and Arunachal Pradesh are separated almost entirely by registration volume (avg. 25.5M vs. 1.2M total registrations), not adoption behaviour direct confirmation that the data-quality issue from Slide 5 is large enough to distort even the unsupervised analysis.

Cluster 4 (Goa, Chandigarh, Kerala) posts the highest average 2024 rate (13.89%) these are regions that started lower and climbed fastest, a genuinely different policy situation from Cluster 0's already-established leaders.

FINAL OUTPUT & RECOMMENDATIONS

Eight Regions, One Consensus Shortlist, Read With Its Caveats

Region	2024 Rate	2024 EVs	Logistic	Rand. Forest
Goa	16.44%	5,754	95.3%	98.4%
Chandigarh	13.87%	2,450	97.8%	97.0%
Tripura	12.29%	3,243	99.7%	96.5%
Delhi	11.99%	34,343	97.9%	94.0%
Kerala	11.35%	35,344	94.0%	85.7%
Karnataka	9.10%	72,539	78.7%	98.2%
Assam	8.65%	25,062	81.1%	96.2%
Uttar Pradesh	8.26%	130,189	64.6%	89.6%

Maharashtra (8.11%) is excluded only because the two models disagree (RF 80.5% vs. Logistic 37.5%) — with 94,000+ EVs registered, it deserves manual review, not automated exclusion.

PLANNING MATRIX: RATE × VOLUME

High Rate · High Volume

Karnataka, UP, Delhi
(Maharashtra: review)

High Rate · Lower Volume

Goa, Chandigarh,
Tripura monitor & target

Lower Rate · High Volume

Watch closely - small %
shifts = large absolute EVs

Lower Rate · Lower Volume

Limited near-term demand
continue monitoring

KEY RECOMMENDATIONS & LIMITATIONS

- Always pair adoption rate with EV volume never rank on rate alone.
- Report graded probabilities, not binary labels; set the decision threshold below 0.5 to favour recall (a missed growing region is costlier than an early charger).
- Clean the 342 anomalous rows before extending this analysis Rajasthan's rank is currently unreliable.
- This is a directional shortlist built on n = 34, with no external validation on a future period not a settled ranking for capital allocation.

IIM KASHIPUR

THANK YOU